

Seyyed Hossein Monsefi, MSc

Computational Fluid Dynamics & Modeling Expert

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Am Langen Felde 46/31, 1220, Wien | Work Authorization: Austria (Red-White-Red Card Plus)

PROFESSIONAL SUMMARY

CFD engineer with 7+ years of experience in fluid dynamics simulations, turbulence modeling, and heat and mass transfer. Proven track record of delivering simulation-driven design improvements, including up to 20% gains in mass transfer efficiency through close collaboration with multidisciplinary teams, and validation against experimental data at over 90% accuracy. Author of 3 peer-reviewed publications and recipient of the Best Junior Presentation Award at ESB 2024. Specialized in ANSYS Fluent, RANS/LES turbulence modeling, dynamic mesh, UDF development, simulation automation, and CFD-based geometry optimization.

SKILLS

- Simulation & CFD Tools:	ANSYS Fluent (Advanced), ANSYS Mechanical, ANSYS Meshing, ICEM CFD, CFD-Post, Tecplot, FSI Simulation
- CAD & Modeling:	SOLIDWORKS, ANSYS DesignModeler
- Programming & Scripting:	MATLAB, Python (working knowledge), C++, ANSYS Fluent UDF development
- Experimental Methods:	Micro-PIV measurements, 3D printing & prototype fabrication
- Documentation & Reporting:	Technical report writing, Conference presentations, Data visualization
- Languages:	English (C1), German (A2, currently improving), Italian (A1)

PROFESSIONAL EXPERIENCE

Project Assistant - CFD Modeling & Optimization

TU Wien, Vienna 08/2022-02/2026

Led a work package in the EU-funded BioMembROS project (European Innovation Council), simulating and optimizing hollow fiber membrane oxygenators in collaboration with international consortium partners.

- Developed the first CFD implementation of a physiologically informed gas transfer model for membrane oxygenators, achieving average prediction errors of 2.5% for O_2 and 10% for CO_2, approximately 2× more accurate than existing models.
- Developed a metaheuristic algorithm, Modified Enhanced Jaya, to fit complex nonlinear simulation data with over 95% accuracy.
- Coupled MATLAB with ANSYS Fluent to automate 250 CFD simulation cases, applying Pareto-based multi-objective optimization to identify optimal hollow fiber arrangements.
- Conducted a DoE-based parametric CFD study across 48 fiber cross-section geometries, from conventional cylindrical to sinusoidal profiles, increasing mass transfer by ~20% while maintaining acceptable blood damage levels.
- Validated CFD models against micro-PIV experimental measurements, achieving over 90% agreement in flow patterns and velocity magnitudes.
- Fabricated in-house microchannel prototypes using 3D printing as a rapid prototyping alternative to molding, enabling experimental validation of CFD models.

CFD Engineer - Cardiovascular Flow & Pump Simulation

Medrik Dynamic Technology (collaboration with FineHeart), Remote 01/2019-01/2021

Developed patient-specific CFD models of implantable cardiac pump systems to evaluate hemodynamic performance, flow behavior, and thrombosis risk across multiple pump designs.

- Reconstructed patient-specific left ventricle geometry from 7 MRI images and implemented custom UDFs in ANSYS Fluent to apply physiologically accurate dynamic mesh motion throughout the cardiac cycle.
- Simulated 6 implantable pump configurations under 3 rotational speed profiles (3,000–6,000 rpm), evaluating wall shear stress, stagnant flow regions, and flow transport behavior.
- Optimized pump casing geometry by identifying a design configuration that eliminated near-zero velocity regions and substantially reduced thrombosis risk inside the casing.
- Evaluated coupling blade geometries and reverse-flow angles to improve apex washing and overall flow performance, achieving flow rates up to 2.8 L/min at 6,000 rpm.
- Simulation results showed good agreement with FineHeart's PIV experimental measurements, confirming the validity of the CFD models across the tested pump configurations.

CFD Engineer - Hydrofoil & Turbine Optimization

Independent Project – University of Tehran, Tehran, Iran 12/2017-08/2018

Developed a fully automated simulation framework for hydrofoil optimization in marine current turbines, integrating parametric geometry generation, CFD analysis, and uncertainty quantification workflows.

- Parameterized hydrofoil geometries using Bézier curves with 13 control points, and developed a custom Genetic Algorithm in MATLAB for geometry optimization.
- Built a fully automated geometry-to-results pipeline coupling MATLAB, ANSYS DesignModeler, ICEM CFD, and ANSYS CFX to evaluate 1,000+ hydrofoil configurations.
- Calculated lift and drag coefficients across multiple angles of attack using XFOIL and ANSYS CFX, integrating results with Blade Element Momentum (BEM) theory to estimate turbine power and torque.
- Connected the simulation framework to an uncertainty quantification (UQ) optimization pipeline, enabling robust design evaluation under variable operating conditions.

EDUCATION

Master of Science in Biomedical Engineering 2018 - 2021

Politecnico di Milano, Milan, Italy GPA: 99/110

Bachelor of Science in Mechanical Engineering 2013 - 2017

University of Tehran, Tehran, Iran GPA: 3.88/4

HONORS AND AWARDS

Journal Cover Feature – Membranes, September 2025
First-author paper selected as cover article of the September 2025 issue.

Best Junior Presentation Award – ESB Meeting, TU Wien, May 2024
Awarded at the ANC-ESB Annual Meeting (European Society of Biomechanics) for CFD-based research on oxygen transport in membrane oxygenators.

LIST OF PUBLICATIONS

A Validated CFD Model for Gas Exchange in Hollow Fiber Membrane Oxygenators: Incorporating the Bohr and Haldane Effects *(First Author) - Membranes, 2025* - DOI: 10.3390/membranes15090268

Optimizing Hollow Fiber Membrane Oxygenators: A Multi-Objective Approach for Improved Gas Exchange and Reduced Blood Damage *(First Author) - Journal of Membrane Science, 2025* - DOI: 10.1016/j.memsci.2025.124228

Effects of Pulsatile Bi-level Ventilation on Flow Characteristics in an Alveolated Duct: A Comparative Study *(Second Author) - Computers in Biology and Medicine, 2025* - DOI: 10.1016/j.compbiomed.2025.110509